



Preliminary Design Review

Financial Aid Impact Study

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What?



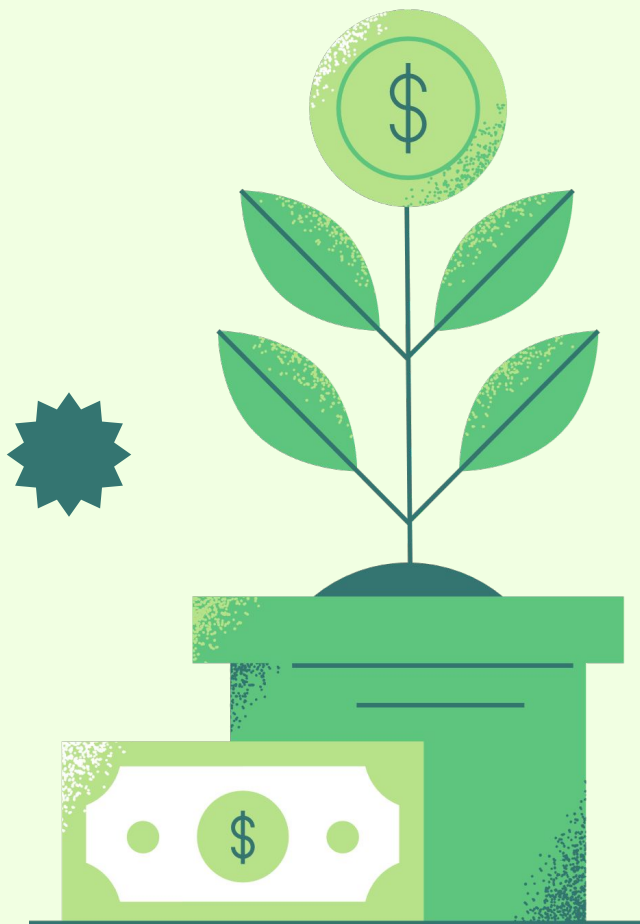
Objective Goal

- Develop a framework that measures the effectiveness of financial aid programs on student success outcomes and their ROI's



Problem

- UCF systematic framework lacks distinguishing financial aid comparison to student success
- A proper analysis needed of programs producing the greatest impact



WHY?

- Helps UCF leadership & staff understand which financial aid programs can contribute most to student success.
- Minimizes unnecessary programs
- Prioritizes the programs that are more statistically beneficial.



HOW?

WHEN?

WHO?



Management Structure

- Starting off with a fairly clean dataset with no more than a couple variables
- Can add as many variables as we can to find our outcome



Budget and Funding Status

- There is no financial budget required for our project
- Time budget of deciding flexibility and priorities to improve efficiency



Risk Management

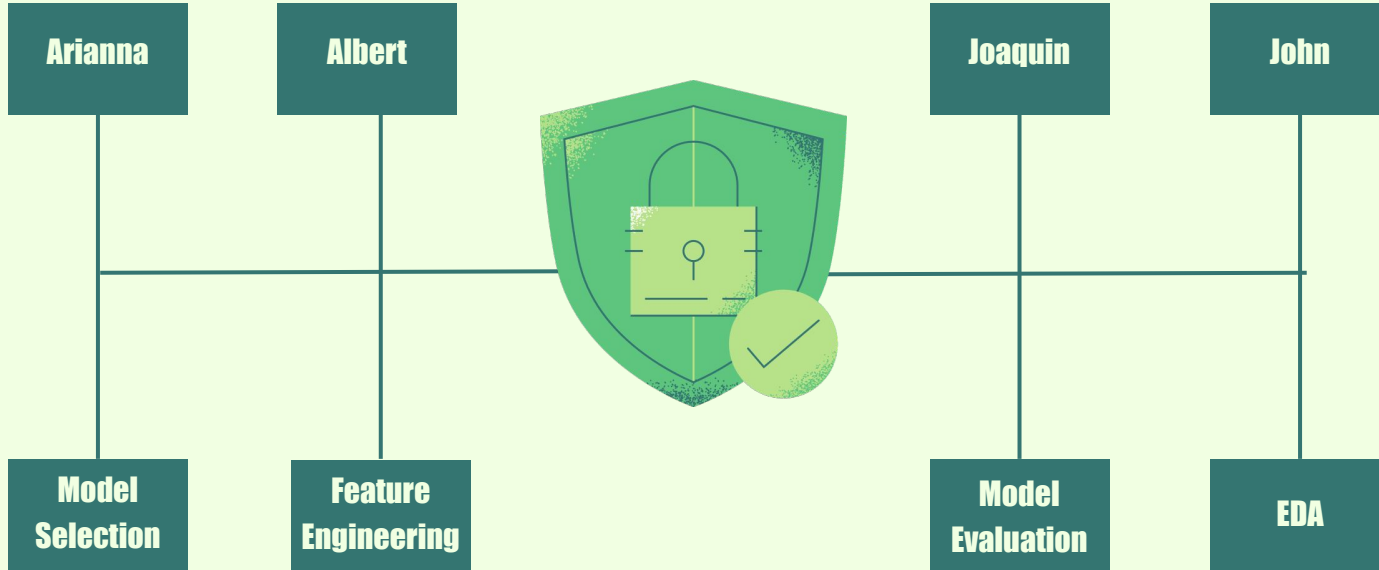
- For our risk management we plan to outline every possible risk and establish the probability of it occurring as well as an impact



Scheduling & Milestones

- For our goals we want to create a milestone that we can reach every week to further our work on assigned tasks/project. Our group will meet at least once a week in order to reach this goal

INFOGRAPHIC

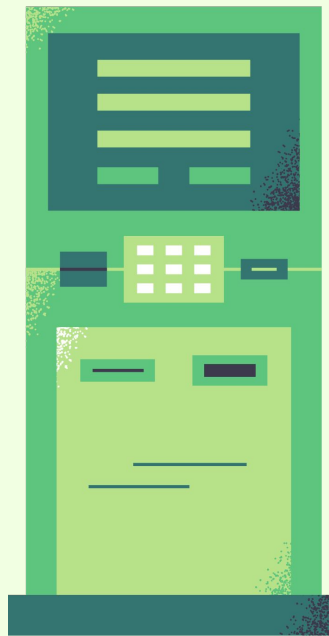




Exploratory Data Analysis

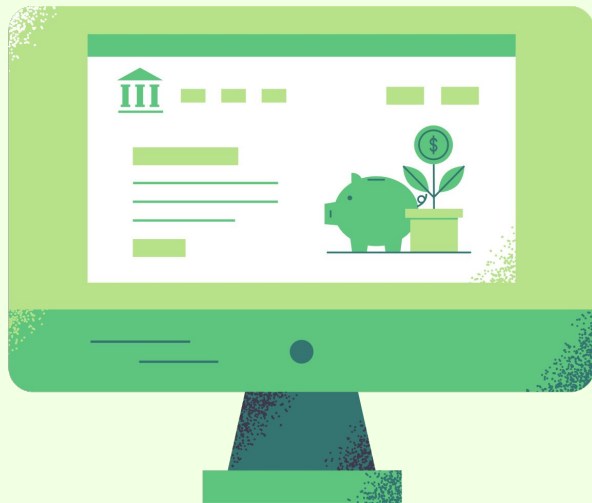


- Examine the distribution of the data and identifying trends, patterns, and possible outliers.
- Verify relationships between variables before applying statistical tests or models.
- Visualizing relationships between variables using graphs.
- Calculate summary/descriptive statistics such as mean, median, standard deviation, and range.
- Check data quality by identifying missing values, duplicates, or unusual observations.



Preprocessing & Feature Engineering

- Importing and inspecting the data through basic sanitation checks
 - Cleaning
 - Removing features containing high percentage of null values
- Feature Engineering
 - Transforming data
 - Creating features
 - Selecting useful features
 - Imputing missing values



Model Selection & Evaluation

- Feature Selection: some columns may include unnecessary data to affect the outcome of our projects
 - Variables include: SAT grade, ACT grade, High School GPA
- Model Evaluation: compare the shortlisted models using validation metrics and choose the best one
- Association model, used for comparing the relationship between variables of our statistical framework
- Double Model Learning methodology to help reduce overfitting bias





Technology & Frameworks



Python Language

Efficient in coding
visualizations of charts
& plots along with easy
numerical computation
tools

R Language

Structured
implementations
making data cleaning
& manipulation
straightforward for
visualizing data



Packages

Both languages have
useful libraries that
are beneficial in D.S.
and M.L. such as
Pandas & Scikit Learn



Risk Assessment

- 1). Incomplete or Inconsistent historical data.
- 2). Delay in creating curated modeling data set.
- 3). Ambiguous assumptions about data and financial aid programs, limiting the validity of analytical conclusions.
- 4). Data governance
- 5). Chosen models failed the evaluation metrics

Catastrophic	1				
Major			3 2	5	
Moderate			4		
Minor					
Negligible					
	Rare	Unlikely	Possible	Likely	Almost Certain

Where?

RISKS	Mitigation
Incomplete or Inconsistent Data	Explore and validate data early, apply relevant data transformations, document and inform project sponsor about data inconsistencies.
Ambiguous assumptions about data and financial aid programs, limiting the validity of analytical conclusions.	Confirm program definitions, eligibility criteria, and assumptions with project sponsor.
Delay in creating curated modeling data set	Prioritize preprocessing tasks, and conduct regular progress reviews to identify bottlenecks and address challenges.
Chosen models failed evaluation metrics	Evaluate baseline and alternative models that satisfy our evaluation criteria.
Data Governance	Follow approved security procedures and restrict sensitive data to prevent data leakage.

Verification & Validation



Model Performance Metrics

Validation:

- 95% Confidence Intervals — Causal ROI Estimates

Regression:

- RMSE
- R^2

Classification:

- F1
- Accuracy

Verify with project stakeholders for our validation



Future Actions



Aggregate all necessary

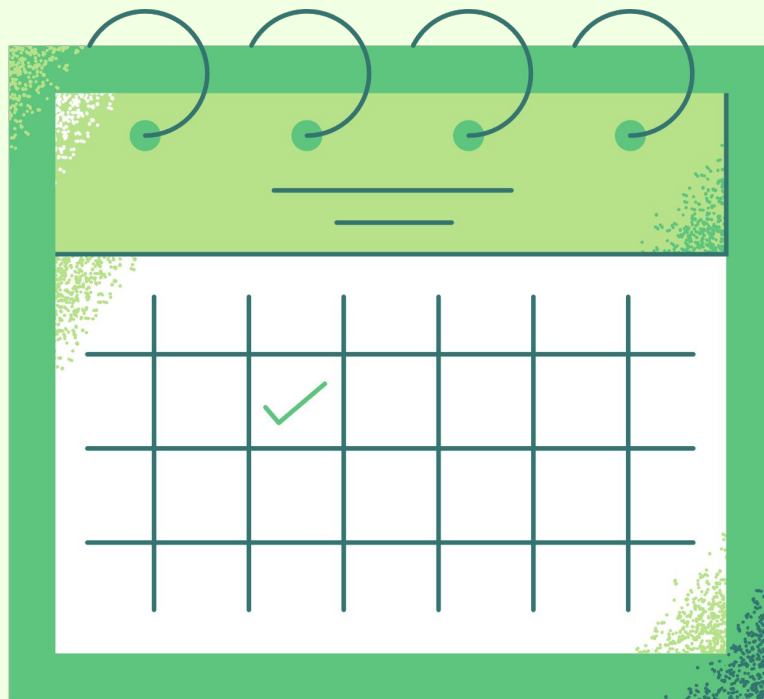
datasets

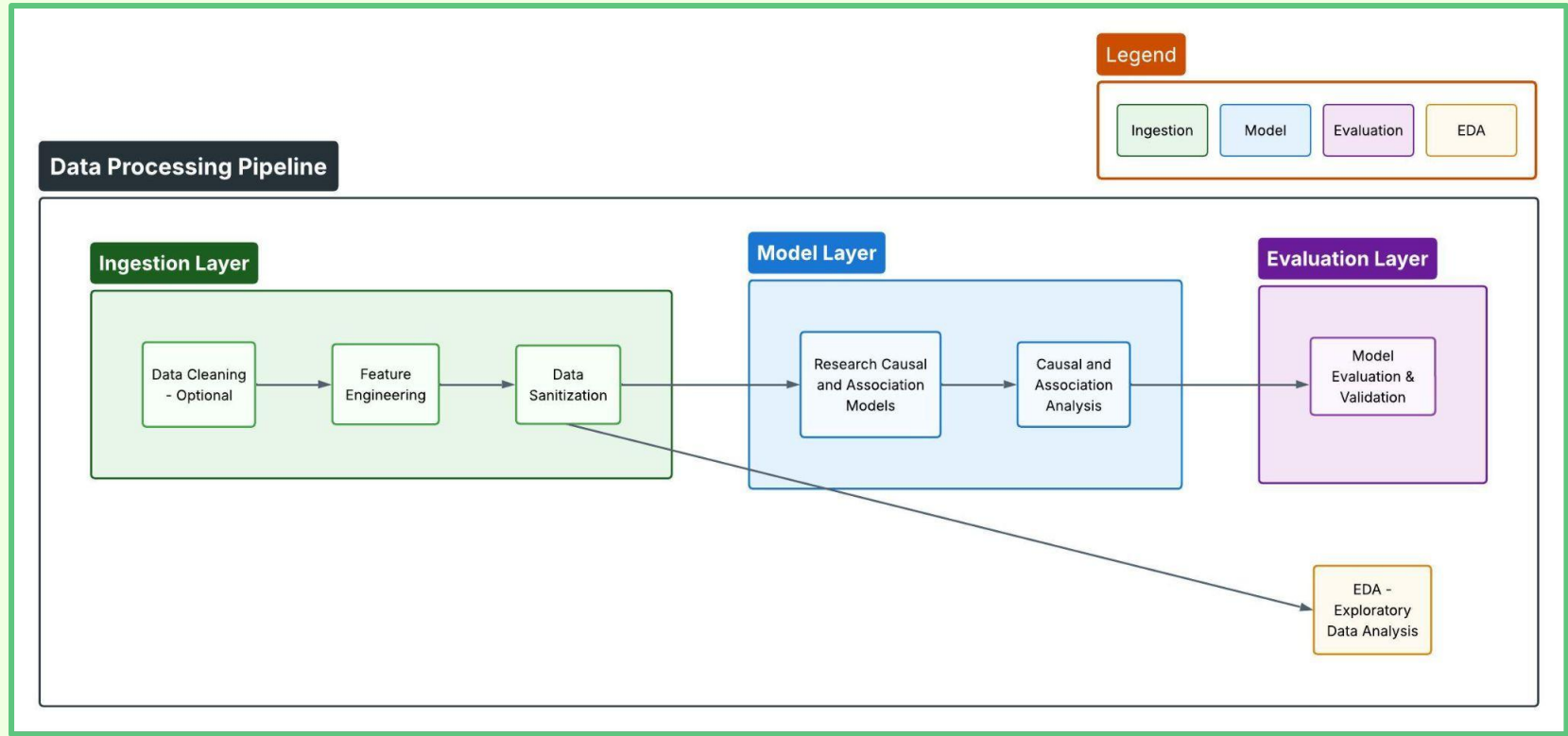


EDA and Feature Engineering



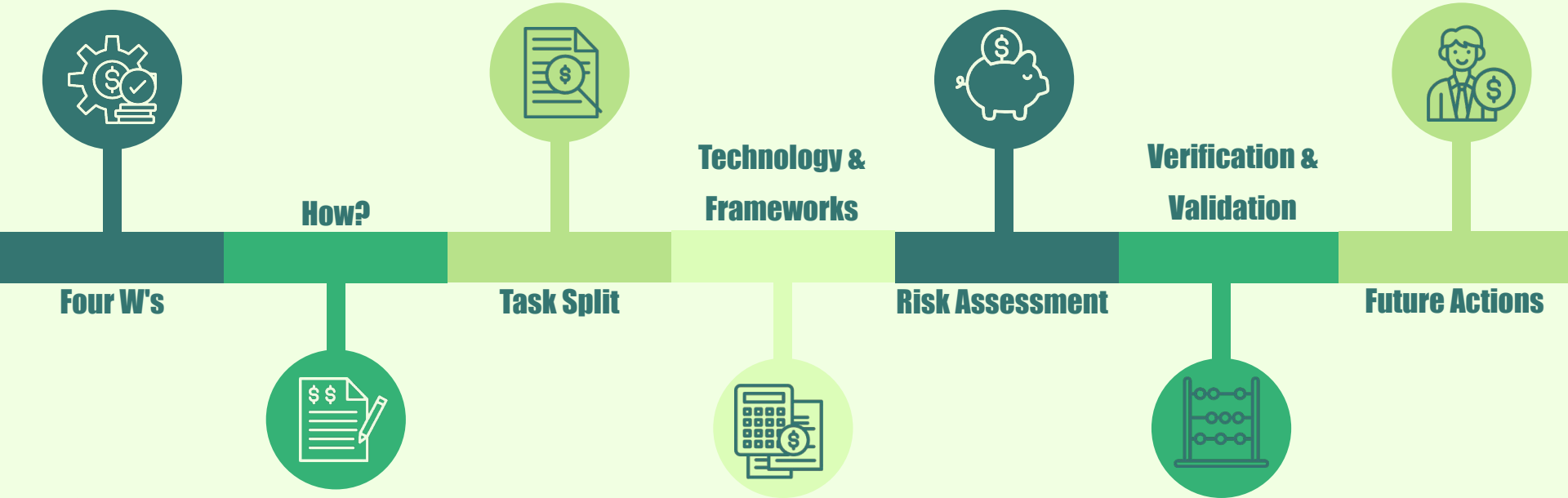
Model Selection & Evaluation





Preliminary Design

Summary



The background of the image is a collage of several US one-dollar bills, slightly overlapping and tilted at various angles. The bills are a light green color with the portrait of George Washington. The text 'THE UNITED STATES OF AMERICA' and 'ONE DOLLAR' are visible on the bills. A semi-transparent grey rectangle is overlaid on the bills, containing the text 'Thank You' and 'Questions?'.

Thank You

Questions?